

EcoTwin Product Requirements Document (PRD)

Ecosystem Analog Discovery and Forecasting Platform

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1 Document Information

Project Name	EcoTwin
Version	1.0
Document Type	Product Requirements Document
Prepared By	Project Team
Target Release	MVP Release

2 Executive Summary

EcoTwin is an ecosystem discovery and forecasting platform that enables users to explore ecosystems using satellite imagery, identify ecologically similar regions, analyze historical environmental trends, and generate future ecosystem forecasts.

The platform leverages Earth Observation data, geospatial foundation models, ecosystem similarity search, and analog forecasting to help researchers, environmental planners, policy-makers, and conservation organizations better understand ecosystem evolution.

The system will provide an intuitive web-based interface where users can interactively select geographic locations, discover ecosystem analogs, visualize environmental changes, and generate automated ecosystem reports.

3 Problem Statement

Environmental ecosystems are continuously changing due to climate variability, urbanization, agricultural expansion, and natural processes.

Current monitoring systems typically focus on descriptive observations and rarely provide mechanisms to:

- Discover ecologically similar ecosystems.
- Compare ecosystem trajectories.
- Learn from historical analog ecosystems.
- Forecast future ecosystem changes.

Researchers and planners often lack tools that can answer:

“What ecosystems elsewhere have evolved similarly to this ecosystem, and what can we learn from their historical evolution?”

EcoTwin aims to address this challenge.

4 Vision

To build an intelligent ecosystem intelligence platform capable of:

- Discovering ecosystem analogs.
- Understanding ecosystem evolution.
- Providing explainable ecosystem forecasts.
- Supporting environmental decision making.

5 Goals

5.1 Business Goals

1. Demonstrate ecosystem analog search.
2. Enable interactive ecosystem exploration.
3. Support environmental forecasting workflows.
4. Provide explainable ecosystem insights.

5.2 Product Goals

1. Allow users to select any location within the study area.
2. Automatically identify the corresponding ecosystem region.
3. Retrieve similar ecosystem regions.
4. Display ecosystem evolution through time.
5. Generate analog-based forecasts.
6. Generate downloadable ecosystem reports.

6 Target Users

6.1 Primary Users

- Environmental Researchers
- Conservation Scientists
- GIS Analysts
- Remote Sensing Researchers
- Ecologists

6.2 Secondary Users

- Policymakers
- NGOs
- Government Agencies
- Academic Institutions
- Students

7 User Personas

7.1 Persona 1: Environmental Researcher

Name: Research Scientist

Needs:

- Discover similar ecosystems.
- Analyze ecosystem evolution.
- Generate scientific insights.

Success Criteria:

- Can retrieve ecosystem analogs.
- Can compare environmental trajectories.

7.2 Persona 2: Conservation Planner

Needs:

- Understand ecosystem risks.
- Forecast future ecosystem conditions.
- Support planning decisions.

Success Criteria:

- Receives explainable forecasts.
- Can generate ecosystem reports.

8 Product Scope

8.1 In Scope

- Interactive ecosystem map.
- Location selection.
- Ecosystem region identification.
- Ecosystem similarity search.
- Historical trend visualization.
- Analog forecasting.
- Automated reporting.

8.2 Out of Scope

- Real-time satellite ingestion.
- Species-level prediction.
- Climate simulation.
- Hydrological simulation.
- Global-scale deployment.

9 User Journey

1. User opens EcoTwin.
2. User navigates the map.
3. User clicks a location.
4. System identifies the ecosystem region.
5. System displays ecosystem summary.
6. User requests similar ecosystems.
7. System retrieves ecosystem analogs.
8. User explores historical trajectories.
9. User generates ecosystem forecast.
10. User downloads report.

10 User Stories

10.1 Location Selection

US-001

As a user,
I want to select a location on the map,
So that I can analyze the ecosystem around that location.

Acceptance Criteria:

- User can click on a map.
- Selected region is highlighted.
- Region metadata is displayed.

10.2 Ecosystem Identification

US-002

As a user,
I want the system to identify ecosystem characteristics,
So that I can understand the selected region.
Acceptance Criteria:

- Ecosystem type displayed.
- Environmental indicators displayed.
- Region summary generated.

10.3 Similarity Search

US-003

As a user,
I want to discover similar ecosystems,
So that I can compare ecological behavior.
Acceptance Criteria:

- Top similar ecosystems returned.
- Similarity score displayed.
- Results shown on map.

10.4 Historical Analysis

US-004

As a user,
I want to view historical ecosystem evolution,
So that I can understand long-term changes.
Acceptance Criteria:

- Historical charts available.
- Multiple years visualized.
- Comparison between ecosystems supported.

10.5 Forecasting

US-005

As a user,
I want ecosystem forecasts,
So that I can anticipate future changes.
Acceptance Criteria:

- Forecast generated.
- Confidence score displayed.
- Forecast explanation provided.

10.6 Report Generation

US-006

As a user,
I want downloadable reports,
So that I can share ecosystem insights.
Acceptance Criteria:

- PDF generated.
- Charts included.
- Analog ecosystems included.
- Forecast included.

11 Functional Requirements

11.1 FR-001

The system shall allow users to select locations on an interactive map.

11.2 FR-002

The system shall identify the ecosystem region associated with the selected location.

11.3 FR-003

The system shall display ecosystem metadata.

11.4 FR-004

The system shall retrieve similar ecosystems.

11.5 FR-005

The system shall display similarity scores.

11.6 FR-006

The system shall display ecosystem analogs on a map.

11.7 FR-007

The system shall visualize historical environmental indicators.

11.8 FR-008

The system shall support comparison of multiple ecosystem trajectories.

11.9 FR-009

The system shall generate ecosystem forecasts.

11.10 FR-010

The system shall generate downloadable reports.

12 Non-Functional Requirements

12.1 Performance

- Region lookup < 1 second
- Similarity search < 2 seconds
- Forecast generation < 5 seconds
- Report generation < 10 seconds

12.2 Scalability

System should support:

- 5,000+ ecosystem regions
- Historical records for multiple years
- Concurrent user access

12.3 Usability

- Intuitive interface
- Minimal learning curve
- Interactive visualizations

13 Success Metrics

13.1 Product Metrics

- Successful location selection rate > 95
- Similarity retrieval success > 90
- Report generation success > 95

13.2 Performance Metrics

- Average search latency < 2 seconds
- Average forecast generation < 5 seconds
- Average page load < 3 seconds

13.3 User Experience Metrics

- User task completion rate > 90
- Positive usability feedback
- Successful ecosystem discovery workflows

14 Risks

Risk	Impact	Mitigation
Medium	Cloud filtering	Satellite Data Quality
Confidence Scores	Forecast Uncertainty	High
Large Data Volume	Medium	Caching and indexing
		Similarity Retrieval
		Quality
High	Embedding validation	

15 Release Plan

15.1 Phase 1

Core Infrastructure

- Map Interface
- Region Selection
- Ecosystem Summary

15.2 Phase 2

Ecosystem Discovery

- Similarity Search
- Analog Retrieval
- Historical Visualization

15.3 Phase 3

Forecasting and Reporting

- Analog Forecasting
- PDF Reports
- Final Optimization

16 Conclusion

EcoTwin will provide a comprehensive ecosystem discovery and forecasting platform that combines ecosystem analog search, temporal analysis, and explainable forecasting into a unified user experience.

The platform will enable researchers, planners, and environmental stakeholders to better understand ecosystem evolution and support evidence-based environmental decision making.